

Arc Length and Improper Integrals

ANSWER KEY



Arc length

- 1 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.
 $y' = x^{1/2}$, so $L = \int_0^3 \sqrt{1+x} \, dx = \left[\frac{2}{3}(1+x)^{3/2} \right]_0^3 = \frac{2}{3}(8-1) = \frac{14}{3}$.
- 2 Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.
 $L = \int_0^\pi \sqrt{1 + \cos^2 x} \, dx$.

Improper integrals

- 3 Evaluate $\int_1^\infty \frac{1}{x^2} \, dx$, or state that it diverges.
 $\lim_{b \rightarrow \infty} \left[-\frac{1}{x} \right]_1^b = \lim_{b \rightarrow \infty} \left(1 - \frac{1}{b} \right) = 1$: converges to 1.
- 4 Evaluate $\int_1^\infty \frac{1}{x} \, dx$, or state that it diverges.
 $\lim_{b \rightarrow \infty} [\ln x]_1^b = \lim_{b \rightarrow \infty} \ln b = \infty$: diverges.
- 5 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} \, dx$ (improper at the lower limit).
 $\lim_{a \rightarrow 0^+} [2\sqrt{x}]_a^1 = 2 - 0 = 2$: converges to 2.